



Software Engineering and Capstone Project: ICS 499-50

Fall 2026: 4 Credits: Online Course

Semester Start Date: 25 August 2026

Semester End Date: 1 December 2026

Instructor Information

Name: Siva Jasthi

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Please note that email is my preferred method of contact and will result in faster responses to you. Be sure to put the course name in the subject line so that I can differentiate between all the different students and other University community members I work with. Remember to use your Metro email account to avoid your message going to the spam folder.

ZOOM

Topic: ICS 499 (8/25 to 12/1)

Time: Aug 25, 2026 06:00 PM Central Time (US and Canada)

Every week on Tue, until Dec 1, 2026, 15 occurrence(s)

<https://minnstate.zoom.us/j/98142421560>

Meeting ID: 981 4242 1560 Passcode: 1201

Department Information:

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About the Course

Course Description

This course focuses on the theory and practice of effectively and efficiently building software systems that satisfy the requirements placed upon them by customers. This course gives an overview of the software lifecycle and introduces various process models used to develop software.

Prerequisites

- [ICS 370: Software Design Models](#) or
- [ICS 372: Object-Oriented Design and Implementation](#)

Competence Statements/Course Learning Outcomes

Competence Statement

This course focuses on the theory and practice of effectively and efficiently building software systems that satisfy the requirements placed upon them by customers. This course gives an overview of the software lifecycle and introduces various process models used to develop software.

In the capstone course, students complete projects, emphasizing the workplace environment that the computer scientist or IT professional will encounter. As often as possible, the projects will be real projects done for real clients. Technical skills and knowledge to support software development challenges facing the information systems professional will be addressed through lectures, readings, and assignments in software engineering. The professional responsibilities of an information systems professional will be examined through readings, assignments, and discussions.

Learning Outcomes

Upon completion of this course, students will:

- Effectively and efficiently build software systems.
- Design and document a complete system and implement it
- Write user manuals
- Present work orally
- Understand Computer Ethics issues
- Elucidate requirements, and use those build a system that fulfills those requirements.
- Manage the software lifecycle.
- Select an appropriate process model for a project.
- Verify and validate a system.
- Manage a software project, including estimating effort, time, and resources needed.
- Work cooperatively in a group
- Create a project plan and execute it
- Elicit and document system requirements

Learning Methods

This course uses a variety of assignments in order to measure student outcomes. In this course, you will be participating in these activities.

Activity 1: Individual Assignments

It is required that each of you have the required proficiency in developing reasonably complex software systems.

You will be given some programming assignments in the initial stages of the course. These assignments may pave the way for bigger **projects** during the final stages of the course.

I reserve the right to change these to group assignments as needed.

Activity 2: Final Projects and Case Studies

The final project is a significant component of this course. These projects typically involve the following components

- AI/LLM Applications
- GitHub Automation (CD/CD projects)
- Database (MySQL) components
- Programming/Server component (PHP or Python or Java)
- Interaction component (Python Flask or Java or HTML/CSS)
- Web Services (Consumption and Development)
- Python Automation
- Mobile Application Development

The final project assigned to you is usually for a community / non-profit organization. These are typically multi-semester projects (you may be starting a new one or enhancing the existing one). You may be required to speak to customers, and non-profit organizations to solicit requirements and to design solutions for them. It is required that you maintain the confidentiality of these customers and their operations. If your project goes into production, your contributions are typically recognized through an MIT License copyright notice.

The project will be completed in several iterations. Each iteration will enhance the artifacts of the project. Each iteration will also contain the peer evaluation as well. The project iterations are presented in the classroom by demonstrating the functionality or walking through the artifacts.

Activity 3: Final Test:

The final test will evaluate the comprehensive understanding of Software Engineering topics.

Text and Supplemental Readings

Textbook and Other Required Reading Materials

No books are needed for this course. You will rely on internet resources to explore the software engineering and programming topics introduced in this class.

Suggested Supplementary Readings, Databases, and Sources

[1] (Web Technologies) HTML, CSS, JavaScript, MySQL at www.w3schools.com

[2] AI and LLMs: <https://www.codecademy.com/catalog/subject/artificial-intelligence>

[3] *Essentials of Software Engineering 5th Edition*

Authors: Frank Tsui (Author), Orlando Karam (Author), Barbara Bernal (Author)

Publisher : Jones & Bartlett Learning ISBN-10 : 1284228991

[4] A Gift of Fire: Social, Legal, and Ethical Issues for Computing Technology (4th edition)

Author: Sara Baase

Publisher : Pearson; 4th edition ISBN-10 : 0132492679

Evaluation Criteria/Assignment Information

Assignments and Grading

Following are the assignments, points assigned, and grade points.

	No	Points	Total
Assignments	2	25 + 50	75
Final Project - Iterations	10	10	100
Final Project -Release	1	100	100
Final Test	1	50	50
Ethics Quiz	1	10	10
Capstone Survey	1	30	30
Attendance for Final Project Iterations	11	10	110
Total Points			475

Grade Scale

Your letter grade will be determined based on the percentage of possible points that you earn during the quarter. The following table relates the percentage/points to a letter grade. I reserve the right to lower the standards (so as to protect the students from unrealistic expectations of the instructor) after the comparative evaluation of the class.

Grade	Percentage
A	94 - 100%
A-	89 - 93.99%
B+	85 - 88.99%
B	82 - 84.99%
B-	80 - 81.99%
C+	75 - 79.99%
C	73 - 74.99%
C-	70 - 72.99%
D	60 - 69.99%

F	0 - 59.9%
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If you have selected S/N grading option, then to receive a satisfactory rating (S), you must earn at least 69% of the possible points. If you wish to elect S/N grading, you must inform the registrar before the second class meeting.

Schedule of Topics and Assignment Due Dates

Here is the tentative schedule of the Software Engineering topics we cover in this course.

Please see the following Weekly Planner for additional details on the learning activities and outcomes. I reserve the right to change the number/type / due date of the assignments based on the course dynamics. The due dates for the assignments / drop boxes are set in the D2L. Please refer to D2L Course Calendar for further details. If there are any changes to the schedule, I will announce those in the class.

Week	Day	SE Topic	AI Topic
1	25-Aug-26	ICS499: Introduction Capstone Readiness Handout	AI in Software Development
2	1-Sep-26	Waterfall Model (SDLC) Agile (Scrum) Model (SDLC)	Prompt and Context Engineering
3	8-Sep-26	Final Project - What are you building?	Extending LLMs Beyond Training Data
4	15-Sep-26	Introduction to SCM and GitHub	RAG: Retrieval-Augmented Generation
5	22-Sep-26	Project Management	AI for Content Creation
6	29-Sep-26	Risk Management	Making your Repo AI/LLM friendly
7	6-Oct-26	Testing Strategies and TDD	LLM Evals: Metrics and Frameworks
8	13-Oct-26	Creative Commons and Open Source Software	Prompts, Instructions, and Skills: What is the difference
9	20-Oct-26	Global Software Development	Context is Currency: Tokens and Costs
10	27-Oct-26	Cloud Computing	What is an AI Agent?
11	3-Nov-26		AI Development Frameworks
12	10-Nov-26		LLM Deployment & Operations
13	17-Nov-26	Recap and Final Project Working Session	
14	24-Nov	Final Project Presentations (Batch 1)	
15	1-Dec	Final Project Presentations (Batch 2)	

Assignments Schedule

Assignment	Final Project	Category	Points	Due Date	Due Time
Assignment 1		Assignments	25	25-Aug-26	11:59 PM CST
				1-Sep-26	11:59 PM CST
Assignment 2		Assignments	50	8-Sep-26	11:59 PM CST
	FP1	Final Project	10	15-Sep-26	11:59 PM CST
	FP2	Final Project	10	22-Sep-26	11:59 PM CST
	FP3	Final Project	10	29-Sep-26	11:59 PM CST
	FP4	Final Project	10	6-Oct-26	11:59 PM CST
	FP5	Final Project	10	13-Oct-26	11:59 PM CST
	FP6	Final Project	10	20-Oct-26	11:59 PM CST
	FP7	Final Project	10	27-Oct-26	11:59 PM CST
Ethics Quiz	FP8	Final Project	10	3-Nov-26	11:59 PM CST
	FP9	Final Project	10	10-Nov-26	11:59 PM CST
	FP10	Final Project	100	17-Nov-26	11:59 PM CST
Final Test (50)	FP-Final	Final Project	100	24-Nov	11:59 PM CST
Capstone Survey			30	1-Dec	11:59 PM CST

Copyrights & Intellectual Property

All rights reserved. This syllabus or class materials (notes, slides, assignments, solutions, examples) may not be reproduced, displayed, modified or distributed without the express prior written permission of the copyright holder.

- The redistribution of audio or video recordings from the course to individuals who are not students in the class is prohibited without the express permission of the faculty member and any of the students who are recorded.
- All material for this course is restricted to members of this course. Material may not be redistributed without the permission of the instructor.
- All the data, documents, software provided by the customers are strictly confidential. Students engaged in such projects agree to transfer the copyrights to the respective customers.

CCourse Policies

Late Policy for Assignments

1. The deadline for each assignment is controlled by D2L. It is mandatory that you submit your assignments by the deadline.
2. Any requests to extend the due date are not entertained as it is not fair to other students who take deadlines and schedules seriously.
3. However, I will entertain late submissions with a late penalty as follows.
 - (a) If the late submission is done within 24 hours (of the due date and time), a late penalty of 25% is imposed.
 - (b) No submission will be accepted beyond this grace period.

Using AI and LLMs:

Generative Artificial Intelligence (GAI) tools, such as claude.ai, ChatGPT, GitHub CoPilot can be used for any assignment with appropriate citation. Examples of citing AI language models are available in a collaborative handout from the Writing Center and Library. Students are responsible for fact checking statements composed by AI language models and for evaluating the validity of sources cited in generated text. Since GAI tools are better suited for some tasks than others, the instructor may discuss best practices and recommended uses in class.

Plagiarism and Copyright

Plagiarism is defined as the act of claiming another person's work as one's own. This can be copying or copying even parts of sentences from an article, journal, internet site, another student's work, or other written work. It can also mean using another student's assignment and making minor changes. Changing a few words around in content is still plagiarism.

Sometimes students are not sure if they are committing plagiarism. Please consult the instructor regarding more information on plagiarism and how to avoid it. The instructor reserves the right to assess through <http://www.turnitin.com/> and/or asking for resources used. Students are encouraged to consult the Metropolitan State University Writing Lab and resources for assistance as well.

_As part of academic honesty, students are expected to respect intellectual property laws, including copyright law. No parts of this site, your textbook, other publisher materials, or contribution from other class members can be used or distributed outside of this class without the permission of the author(s). Copyright law provides that any creative thought fixed into tangible form is copyrighted the moment it is created, whether or not a copyright notice is affixed.

Please review this excellent article on plagiarism to learn more. Parts of the article are posted below: <https://www.plagiarism.org/article/what-is-plagiarism>

Acts of plagiarism may be reported to the Associate Provost for Student Success.

Attendance

Your submission of the assignments is used to track your attendance.

Note: Federal Financial Aid regulations require that the University track ongoing attendance.

University Non-Attendance and Reporting Policy and Procedure

The purpose of the Non-Attendance and Reporting Policy is to ensure Federal Title IV regulations are adhered to with respect to a student's enrollment level for the purpose of calculating and paying financial aid. While Metro State University is not required to take attendance, Federal Title IV financial aid regulations require a procedure to establish that students have attended, at a minimum, one day of class for each course in which the student's enrollment status was used to determine eligibility for the Pell Grant Program. In addition, the University needs to determine a last date of attendance for those students who receive all failing grades or unofficially withdraw. Attendance is defined based on course delivery mode. A student is "in attendance" if the student meets the following conditions before the end of the second week of the course:

Classroom Courses –the student is present in the classroom.

Web-Enhanced (Reduced Seat Time Courses) –the student is present in the classroom or submits at least one academically relevant assignment.

Online Courses –the student submits at least one academically relevant assignment.

Independent Studies – the student contacts the instructor or submits at least one academically relevant assignment.

If a student does not attend the first two classes, either live and/or online, that student is automatically dropped from the course. If a student adds the course past the drop/add date, the student will not receive points for any assignments, discussions, quizzes, or exams for which the due date has already occurred. Refer to the Non-Attendance and Reporting Policy 2259, and the Non-Attendance and Reporting Procedure 259.

Note to the student: The above description is the University Policy, but for some courses, based on how it is listed in the registration materials, participation must occur during the first week of class or the student is dropped from the course.

Center for Accessibility Resources

Diversity and Disability Statement: Our institution values diversity and inclusion; we are committed to a climate of mutual respect and full participation. Our goal is to create learning environments that are usable, equitable, inclusive and welcoming. If there are aspects of the instruction or design of this course that result in barriers to your inclusion or accurate assessment or achievement, please notify the instructor as soon as possible. Students with disabilities are also welcome to contact the Center for Accessibility Resources to discuss a range of options to removing barriers in the course, including accommodations.

The Center for Accessibility Resources is located in New Main, room L223. Phone number is 651-793-1549 and email is Accessibility.Resources@metrostate.edu.

Other Student Support Resources

Academic Success Services

Information about academic success services such as library support, tutoring, advising, career services, and other academic success supports can be found on the [Academic Success Services](#) webpage.

Basic Needs, Services, and Support

Information about Metro and community support services for basic needs like housing, food, childcare, and healthcare, as well as information about counseling services, and student community support can be found on the [Basic Needs, Services, and Support](#) webpage.